

Course: B. Pharmacy

Semester: V

Subject Name: Medicinal Chemistry-II

Subject Code: BP501T

Max Marks: 75

Date: 19/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) Explain briefly mechanism of action of alkylating agents
- ii) Mention the structures and uses of amlodipine and nicardipine.
- iii) Write the structure and uses of synthetic estrogens.
- iv) Write briefly about nitro vasodilators
- v) Write the structure and uses of menadione and clopidogrel.
- vi) What are glucosidase inhibitors? Give example with structure
- vii) Enlist the insulin preparations
- viii) Give the structure and uses of mannitol
- ix) Outline synthesis of Acetazolamide
- x) Write any two structures of anilide derivatives used as local anesthetic.

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) What are antihistaminic agents? Classify them with suitable examples and write the synthesis of diphenhydramine HCl
- ii) Classify local anaesthetic agents and explain their SAR and mechanism of action. outline the synthesis procaine.
- iii) Define and Classify antineoplastic agents with suitable examples. Explain the synthesis and uses of Methotrexate.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Explain the chemistry of cardiac glycosides used in CHF.
- ii) Describe the stereochemistry of steroids.
- iii) Define anticoagulants and give the mechanism of action and synthesis of Warfarin
- iv) What are hypoglycemic agents? Describe structural activity relationship of sulfonylureas.
- v) Write a note on gastric proton pump inhibitors.
- vi) Write a note on antithyroid drugs and give the structure and uses of any two.
- vii) What are potassium sparing diuretics? Outline the synthesis and medicinal uses Chlorthiazide.
- viii) Discuss mechanism of action of drugs used for erectile dysfunction and give any two structures and uses.
- ix) Classify antianginal agents and outline the synthesis of nitroglycerine.

*** END OF THE PAPER ***